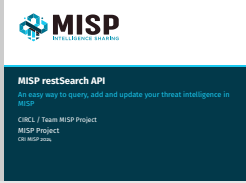
MISP restSearch API

An easy way to query, add and update your threat intelligence in MISP

CIRCL / Team MISP Project

MISP Project

CRI MISP 2024



MISP API reworked

- The MISP API has grown gradually with a UI first design in many cases
- Endpoints all solved specific issues with their own rulesets
- Growth was organic - whenever the need to add a new functionality / filter popped up we've added it
- Lead to frankenmonsters such as this:

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Goals we've set for ourselves

- Open up every functionality in MISP available via the UI to the API
- Including ones related to **instance management**
- APIs that expect input objects for data creation should be **self-describing**
- **URL parameters should be discouraged**, but still usable by legacy tools (deprecation)
- APIs should be heavily **tested** (Raphael Vinot's exhaustive test suite in PyMISP)
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Export API's reimagined

- Scrapped all existing type specific APIs (**deprecated**, documentation moved to legacy, still available)
- **Single entry point** - all export APIs baked into restSearch
- Queries consist of a combination of:
  - **Scope** (Event, Attribute, Sighting, more coming in the future)
  - **Filter parameters** - passed via JSON objects, url parameters (key value or ordered list)
  - A **return format**
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- Under the hood now also used by the UI search and exports

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  - **Scope specific** pagination (number of events, attributes, etc)
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Performance tuning

- Single execution with subqueries
- Internal pagination **aligned with memory limits**
  - Probing of available memory for the current process
  - **Chunking of the query results** to fit in object specific memory envelopes
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- The result of our own frustration
- Built in **ReST client** with templating
- Extensive query builder UI by Sami Mokaddem
- Build queries in a simple interface, automatically set URLs, headers, etc
- Uses the self documentation of APIs
- Export your queries as **cURL or Python scripts**
- Built in testing tools (performance measurements, result parsers)
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- The **maturity** of the communities and threat intel sharing at large has improved
- We are sharing more
- Most importantly: we are sharing **more context** along with technical indicators
- This allows us to **manage our data more accuractely** before feeding them to our protective tools
- Different contexts (APT targeting me? Persisting techniques?) - lifecycle management
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CLI tools for the CLI God, automation for the automation throne

- Open up commonly used system management tasks to the CLI
  - sync servers/feeds
  - caching feeds
  - Password resets
  - Server settings
  - Bruteforce protection resets
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- Add export modules to the restSearch API
- Improve the query language to support some missing features (such as AND boolean operators)
- Support for extended events via the restSearch API
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  - Proof of concept of the system implemented by Christian Studer already works using the STIX / STIX2 export subsystems
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